

MATH*4600
Generative Score-Based Models: Reverse Engineering
Noise

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Introduction

Score matching provides a novel approach to generative modeling by directly estimating the gradients of the data distribution’s log probability density [1, 2]. This technique has shown a marked improvement over other generative modeling techniques such as likelihood-based models and implicit generative models [3, 4].

Likelihood-based models learn the probability density of distribution directly by estimating the maximum likelihood. Examples include auto-regressive models [5], variational auto-encoders [6], normalizing flow models [7], etc. Implicit generative models, on the other hand, implicitly represent the probability distribution via a model of its sampling process. Generative adversarial networks are a well-known example of this.

Score matching circumvents the challenges of explicit density estimation while maintaining the ability to sample from the learned distribution, this is done via modeling the Stein score function [8], which is simply, the *gradient of the log probability density function*.

Let $f_\theta(x) \in \mathbb{R}$ be a real-valued function whose p.d.f can be defined as:

$$p_\theta(x) = \frac{e^{-f_\theta(x)}}{Z_\theta} \quad (1)$$

where $Z_\theta > 0$ is a normalizing constant dependent on θ , such that $\int p_\theta(x)dx = 1$ and $f_\theta(x)$ is often called an un-normalized probabilistic model. Typically we can train $p_\theta(x)$ by maximizing the log-likelihood of the data:

$$\max_{\theta} \sum_{i=1}^N \log p_\theta(x_i) \quad (2)$$

Z_θ is usually an intractable quantity for any general $f_\theta(x)$, this makes it infeasible to train $p_\theta(x)$ via maximum likelihood, as 2 requires $p_\theta(x)$ to be a normalized probability density function. A model for the score function of a distribution $p(x)$ is learned such that $s_\theta(x) \approx \nabla_x \log p(x)$ and can be parameterized without dealing with the normalizing constant. For instance we can parameterize a score-based model with the model defined in 1 via

$$s_\theta = \nabla_x \log p_\theta(x) = -\nabla_x f_\theta(x) - \underbrace{\nabla_x \log Z_\theta}_{=0} = -\nabla_x f_\theta(x)$$

where the score-based model $s_\theta(x)$ is independent of the normalizing constant Z_θ .

Score-based models can be trained by minimizing the Fisher divergence between the model and the data distributions, where the Fisher divergence is defined as:

$$\mathbb{E}_{p(x)}[\|\nabla_x \log p(x) - s_\theta(x)\|_2^2]$$

Score-matching methods allow you to train models by minimizing Fisher divergence without needing to know the true data score function [9, 10]. Like maximum likelihood training, these objectives can be estimated from data and optimized using stochastic gradient descent. This provides a direct way to train score-based models without requiring adversarial training approaches, which can lead to instability [4] and mode collapse [3]. In essence, by training a

flexible score-based model we can represent a probability distribution by modeling its score function via score matching techniques.

We can then use Langevin Dynamics [11] to draw samples from a trained score-based model $s_\theta(x) \approx \nabla_x \log p(x)$. To accurately estimate scores in regions of low density, [1] suggests perturbing the data with noise at multiple noise scales, σ_i , and training score-based models on the noisy data. The score-function, $\nabla_x \log p_{\sigma_i}(x)$ of noise-perturbed distributions can be estimated by training a Noise Conditional Score-Based Model with score matching such that $s_\theta(x, i) \approx \nabla_x \log p_{\sigma_i}(x)$ for all $i = 1, 2, \dots, L$. As the data distribution is perturbed with continuously growing levels of noise, the perturbation procedure can be represented with stochastic differential equations (SDEs). SDEs, generally, are of the form:

$$dx = f(x, t)dt + g(t)dw \quad (3)$$

where $f(\cdot, t) : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is the drift coefficient, $g(t) \in \mathbb{R}$ is the diffusion coefficient and dw is the infinitesimal white noise via a standard Brownian motion. The perturbation process by can be reversed by using the corresponding reverse SDE of 3, given by:

$$dx = [f(x, t)dt - g^2(t)\nabla_x \log p_t(x)]dt + g(t)dw \quad (4)$$

where dt is the negative infinitesimal time step [12].

This is analogous to reversing the perturbation process with annealed Langevin dynamics [13]. Reverse SDEs can be solved by training a Time-Dependent Score-Based model $s_\theta(x, t)$ to estimate the score function, $\nabla_x \log p_t(x)$, from 4. There are many numerical SDE solvers that can be implemented to solve the reverse SDE for sample generation such as the Euler-Maruyama method or stochastic Runge-Kutta method, etc. These samplers can also be modified to compute exact log-likelihood of score-based generative models [2].

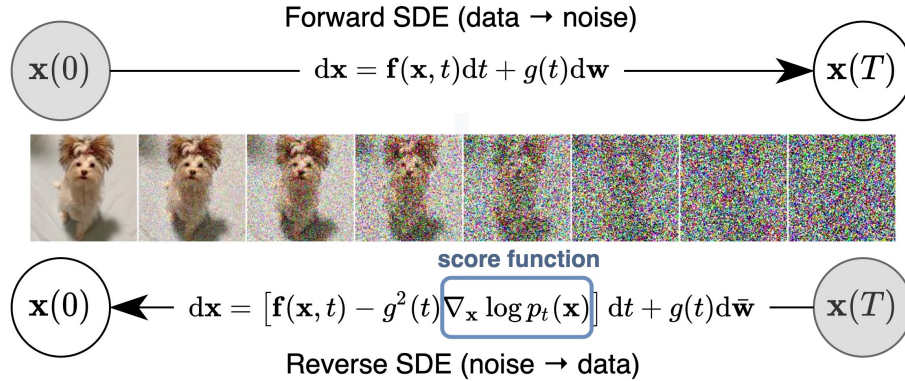


Figure 1: Data can be transformed to a simple noise distribution can be via an SDE, which then can be reversed to generate samples from noise $x(T)$ if we know the score of the distribution at each intermediate time step [1].

Objectives

The primary objective of this research project is to understand, implement and expand on score-based generative models. This will involve the following:

- Conducting experiments to evaluate the performance and capabilities of these generative models across various domains.
- Investigating novel approaches to data distribution perturbation.
- Demonstrating the efficacy of these models to sample from complex distributions that traditional sampling techniques find challenging to navigate.
- Employing score-based generative models to solve inverse problems and apply them to complex tasks, eg: cryptograms.

Timeline

The following is the proposed timeline for this research project:

- Weeks 1-2 (Jan 6-18): Literature review, research proposal, initial low-level implementations of score-based models for simple distributions.
- Week 3 (Jan 19 - 25): More complex implementation of score-based models (eg: using neural networks), replicate findings from [1].
- Week 4 (Jan 26 - Feb 1): Score-based models for challenging distributions beyond MCMC
- Week 5 (Feb 2 - 8): Explore SDEs that transform the data distribution beyond normal distributions.
- Week 6 (Feb 9 - 15): Study the theoretical and practical implications of newly proposed SDE approaches.
- Week 7 (Feb 16 - 22): Validate the proposed SDE modification through computational experiments.
- Week 8 (Feb 23 - Mar 1): Performance analysis, compare novel SDE approach with existing methods.
- Week 9 (Mar 2 - 8): Inverse problem applications, discrete diffusion process literature review.
- Week 10 - 11 (Mar 9 - 22): Application of score-based models to discrete data distribution problems.
- Week 12 (Mar 23 - 29): Refine experimental results, begin drafting research paper, prepare for presentation.
- Week 13 (Mar 24 - Apr 4): Presentation week, writing report.
- Week 14 (Apr 5 - 11): Final report due April 11

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